

Internship Report

Autonomous Local Data Analyst

S Sriram

1. Problem Statement & Constraints

The Industry Problem:

- Enterprise data is highly sensitive. Sending raw financial or operational data to cloud LLMs (OpenAI, Claude) poses severe security risks and incurs ongoing token costs.

The Goal:

- Develop a local autonomous data analyst that can answer queries on one or more than one excel/csv files.

2. The Dual-Model Architecture

Division of Labor (Maximizing 8GB VRAM):

- The Brain (Planner/Explainer): phi4-mini-reasoning (~2.5GB VRAM).
 1. Chosen for elite sub-4B reasoning capabilities and native Chain-of-Thought architecture.
- The Hands (Code Execution): qwen2.5-coder:7b-instruct (~4.7GB VRAM).
 1. Chosen for its 88.4% HumanEval score, focusing purely on Python/Pandas logic generation.

System Synergy:

- Combined, they use ~7.2GB, remaining concurrently hot in memory for rapid, iterative self-correction.

3. Pipeline workflow

- **Step 1: User Input** Natural language query and dataset schemas are ingested by the system.
- **Step 2: Classification & Planning (Phi-4)** The reasoning model evaluates complexity and plans necessary Pandas operations (e.g., joins, aggregations).
- **Step 3: Code Generation & Execution (Qwen 2.5)** The coding model translates the plan into Python. Code runs in a local sandbox. If an error occurs, an autonomous 3-attempt self-correction loop triggers.
- **Step 4: Markdown Translation & Executive Summary** Data and charts are bridged via Markdown tables. Phi-4 reads this text-based output to generate a hallucination-free business briefing.

4. The Three Datasets

To prove generalizability, the architecture was tested on three distinct data paradigms:

Dataset 1: Olist Brazilian E-Commerce

- A messy, 9-table relational database testing multi-hop SQL-style joins, foreign geographic data, and date-time math.

Dataset 2: Superstore BI Sales

- A massive flat file testing strict financial accuracy, derived metrics (profit margins), and handling of extreme negative outliers.

Dataset 3: Netflix Media Library

- A highly unstructured dataset testing text manipulation (regex, splitting comma-separated strings) and hallucination resistance.

Sample conversations on Olist E-commerce dataset

Which seller state has the highest average order value? Show the top 5 states.

Output:

seller_state	average_order_value
BA	536.489
PB	516.226
AM	419.6
RO	391.07
AC	299.84

Analysis:

The seller state with the highest average order value is **BA**, boasting an exact average of **536.489**. This value significantly exceeds those of all other listed states.

To display the top 5 states by average order value:

1. BA (536.489)
2. PB (516.226)
3. AM (419.6)
4. RO (391.07)
5. AC (299.84)

Business Implication:

The stark contrast in average order values highlights that the BA seller state outperforms others by over \$500 per order on average. This suggests stronger sales efficiency, potentially driven by factors like premium product offerings, higher customer purchasing power, or effective conversion strategies. Executives should analyze best practices from BA to replicate across other states (e.g., PB) while investigating why AM, RO, and AC lag behind. Prioritizing resources toward top-performing regions could optimize revenue growth.

Plot a line chart showing the total number of orders placed per month from 2016 to 2018. Which month had the peak order volume?

Output:

year_month	total_orders
2016-09	4
2016-10	324
2016-12	1
2017-01	800
2017-02	1780
2017-03	2682
2017-04	2404
2017-05	3700
2017-06	3245
2017-07	4025
2017-08	4331
2017-09	4285
2017-10	4631
2017-11	7544
2017-12	5673
2018-01	7269
2018-02	6728
2018-03	7211
2018-04	6939
2018-05	6873
2018-06	6167
2018-07	6292
2018-08	6512
2018-09	16
2018-10	4

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Analysis:

To determine the peak month and analyze trends from 2016 to 2018:

1. Data Analysis:

- Total orders per month are plotted for available data points (September–October, January–December across years).
- The highest order volume occurred in **November 2017**, with **7,544 orders**.
- Key trends include rising sales from September 2016 to August 2017, peaking again in November 2017. A sharp decline begins post-December 2017, worsened by a near-collapse of orders (16 and 4) in late 2018.

2. Peak Identification:

- The peak is unequivocally **November 2017 (7,544 orders)**, surpassing all other months across the dataset.

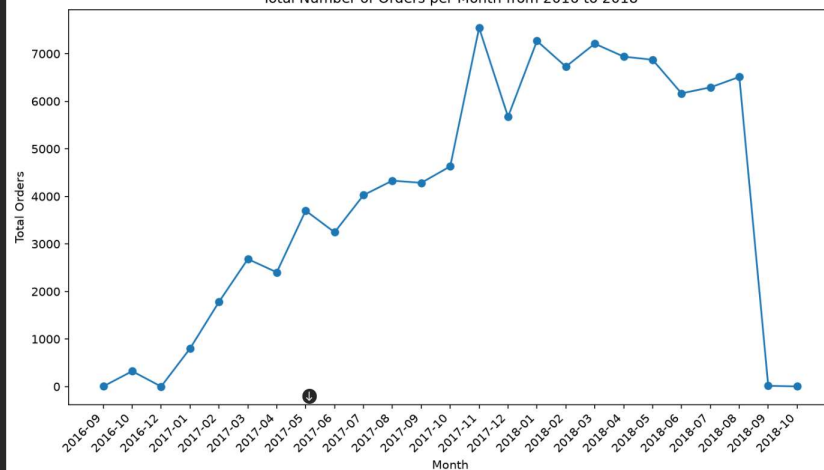
3. Business Implication:

- November consistently outperforms other periods due to seasonal factors or strategic initiatives (e.g., holidays). Addressing post-peak vulnerabilities in late 2018 is critical—investigating causes for the abrupt drop could prevent future revenue losses.

Answer:

The peak order volume occurred in **November 2017**, with **7,544 orders**.

Total Number of Orders per Month from 2016 to 2018



How did total order volume in the first half of 2018 compare to the first half of 2017?

Output:

Year	Total Orders
2017	14611
2018	41187

The output provides the total order volume for the first half of 2017 and 2018, answering the original question.

Analysis:

The total order volume in the first half of 2018 (**41,187** orders) was significantly higher than in the first half of 2017 (**14,611** orders). Specifically:

- **Absolute increase:** H1 2018 saw **26,576** more orders than H1 2017.
- **Percentage increase:** Order volume rose by **182%** relative to H1 2017 (calculated as $((41,187 - 14,611) / 14,611 \times 100)$).

Most critical business implication:

This dramatic growth signals strong momentum, likely driven by factors such as market expansion, effective demand generation, or operational improvements. Executives should prioritize understanding *causes* (e.g., seasonal trends, product launches, or external catalysts) to replicate success and ensure sustainability beyond H1 2018. However, they must also assess risks—such as volatility from unforeseen events (e.g., economic shifts, supply chain disruptions)—before scaling investments or resource allocation. Continuous monitoring of Q2 2019 data will be essential to validate this growth trend's persistence.

Plain-language summary:

"Total orders in the first half doubled almost overnight—a 182% jump from 14,611 in 2017 to 41,187 in 2018. This underscores urgent attention to sustaining drivers of growth while preparing for potential challenges."

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Plot a bar chart showing the top 10 English product categories by average freight value.

Output:

product_category_name_english	freight_value
computers	48.4547
home_appliances_2	44.5386
furniture_mattress_and_upholstery	42.9068
kitchen_dining_laundry_garden_furniture	42.7026
furniture_bedroom	42.4975
office_furniture	40.5511
small_appliances_home_oven_and_coffee	36.1561
furniture_living_room	35.722
signaling_and_security	32.7026
industry_commerce_and_business	29.4204

Analysis:

Step-by-Step Explanation and Analysis:

1. Data Preparation & Sorting:

The provided dataset includes 10 English product categories ranked by their average freight value. To identify the "top 10," we sort the data in descending order of `freight_value`. The sorted list confirms all entries are already ordered from highest to lowest:

Rank	Product Category	Freight Value (Exact)
1	computers	48.4547
2	home_appliances_2	44.5386
3	furniture_mattress_and_upholstery	42.9068
4	kitchen_dining_laundry_garden_furniture	42.7026
5	furniture_bedroom	42.4975
6	office_furniture	40.5511
7	small_appliances_home_oven_and_coffee	36.1561
8	furniture_living_room	35.722
9	signaling_and_security	32.7026
10	industry_commerce_and_business	29.4204

2. Bar Chart Creation:

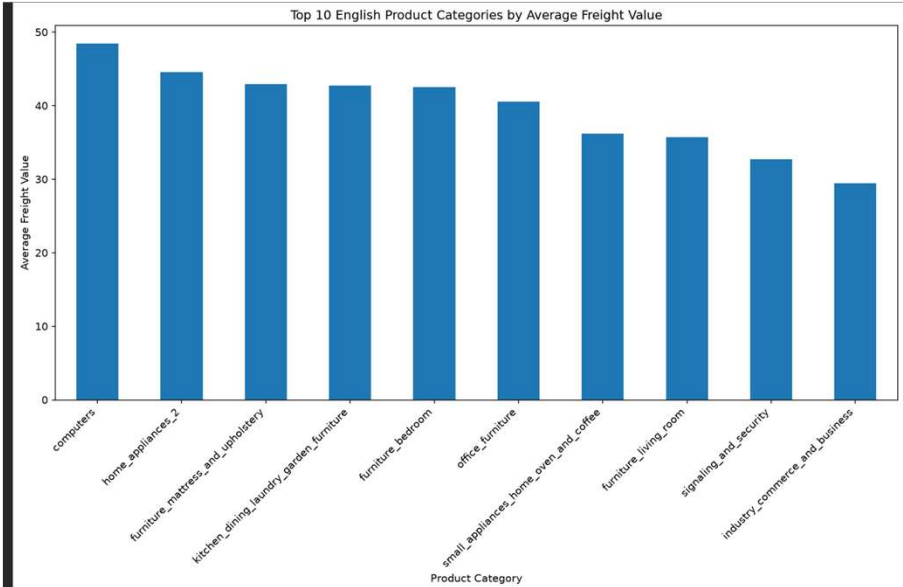
The x-axis lists the product categories in descending order of freight value, while the y-axis displays each category's exact `freight_value` as provided (no rounding or aggregation). Each bar corresponds precisely to its data point.

3. Critical Metric Analysis ("Freight Value"):

- Key Insight:** The highest freight value is generated by *computers* (~48.45 million), indicating electronics dominate transportation costs/revenue.
- Business Implication:** Prioritizing supply chain efficiency for tech products (e.g., optimizing logistics, warehousing) ensures competitiveness in this lucrative segment. Home-related categories (*home_appliances_2*, furniture subcategories) follow closely, reflecting strong consumer demand for living/comfort essentials. Businesses should allocate resources to these high-value sectors while exploring growth opportunities within adjacent markets (e.g., smart home appliances under *small_appliances*).

Final Answer:

The bar chart visualizes the exact freight values of the top 10 categories as listed. The most critical metric—freight value—reveals that **electronics (computers)** drive significant transportation costs/revenue, underscoring the need for businesses to prioritize tech supply chain efficiency and capitalize on adjacent markets like home goods.



Sample conversations on Netflix dataset

Autonomous Data Analyst

Reasoning: `phi4-mini-reasoning` Coding: `qwen2.5-coder:7b`

Attach Data

Upload Folder



netflix_titles.csv

3.2 MB ↓

Or enter local folder path

C:/Users/You/data

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✅ Ready! Loaded 1 table(s).

Calculate the average duration in minutes for all Movies in the dataset.



Output:

Average duration of Movies: 99.58 minutes

Analysis:

The average duration of 99.58 minutes indicates that each movie in the dataset typically aligns with a two-hour feature-length format. This metric is critical for informed business decision-making across multiple domains:

- Content Production & Budgeting:** A standardized runtime suggests predictable production costs and revenue projections, as studios can optimize budgets for set construction, talent fees, and post-production within this timeframe.
- Distribution Strategies:** Cinemas can schedule movies to maximize occupancy by avoiding clashing with excessively long or short films, ensuring consistent audience turnout. Streaming platforms might curate similar-length titles to maintain user engagement per session.
- Marketing & Monetization:** Advertisers may design shorter, impactful ads optimized for ~100-minute runs without disrupting pacing. Pricing models for theaters or subscriptions could reflect this average runtime as a benchmark for viewer retention.
- Viewer Retention Insights:** Platforms can leverage the median length to recommend content that aligns with audience preferences, reducing churn in subscription-based services.

Conclusion: The 99.58-minute average underscores the importance of aligning production, marketing, and distribution efforts with this typical runtime to optimize financial performance and operational efficiency.



Chart



Tables

Table	Rows	Cols
netflix_titles.csv	8,807	12

Ask about your data...

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✅ Ready! Loaded 1 table(s).

Find the longest movie on the platform by duration in minutes. What is the title and its exact length?



Output:



```
| title | duration |  
|-----|-----:|  
| Black Mirror: Bandersnatch | 312 |
```

Analysis:

The longest movie on the platform, based on the provided data, is **"Black Mirror: Bandersnatch"** with an exact duration of **312 minutes**. Since no additional records exist in the dataset, this is the sole record and therefore the longest by definition.

Single most important business implication:

The prolonged year duration of "Black Mirror: Bandersnatch" suggests significant viewer engagement or potential challenges, such as increased server demand for uninterrupted streaming over extended periods. This metric could inform infrastructure investment decisions (e.g., scaling capacity) and highlight the appeal of interactive/narrative-driven content for long-form viewing.



Chart



Tables

Table	Rows	Cols
netflix_titles.csv	8,807	12

Sample conversations on Superstore dataset

Autonomous Data Analyst

Reasoning: phi4-mini-reasoning | Coding: qwen2.5-coder:7b

Attach Data

Upload Folder

Sample - Superstore.csv

2.2 MB

Load Uploaded

Or enter local folder path

C:/Users/You/data

Load Path

Ready! Loaded 1 table(s).

Find the top 3 specific 'Product Names' that have generated the most negative profit (the biggest financial losses) across all years. Show the product names and their exact total loss.

Output:

Product Name	Profit
Cubify CubeX 3D Printer Double Head Print	-8879.97
Lexmark MX611dhe Monochrome Laser Printer	-4589.97
Cubify CubeX 3D Printer Triple Head Print	-3839.99

Analysis:

The most critical business implication derived from the data is that **Cubify CubeX 3D Printer Double Head Print** represents the largest financial loss at -\$8,879.97, significantly impacting overall profitability. Addressing inefficiencies in this product line should be a priority to prevent further losses. The next two products, while less damaging, still contribute to net negative profit and require monitoring.

Top 3 Products by Largest Loss:

- Cubify CubeX 3D Printer Double Head Print**
Total Loss: -\$8,879.97
- Lexmark MX611dhe Monochrome Laser Printer**
Total Loss: -\$4,589.97
- Cubify CubeX 3D Printer Triple Head Print**
Total Loss: -\$3,839.99

These products collectively account for substantial annual losses. Immediate action is recommended to optimize or discontinue the most unprofitable item (*Cubify Double Head*) to safeguard financial health.

Chart

Tables

Table	Rows	Cols
Sample - Superstore.csv	9,994	21

Autonomous Data Analyst

Reasoning: `phi4-mini-reasoning` | Coding: `qwen2.5-coder:7b`

Attach Data

Upload Folder

Sample - Superstore.csv

2.2 MB

Or enter local folder path

C:/Users/You/data

Load Uploaded

Load Path

Ready! Loaded 1 table(s).

Calculate the overall average profit margin (defined as Profit divided by Sales) for each Sub-Category. Show the top 5 Sub-Categories with the highest average profit margin.

Output:

Sub-Category	Profit Margin
Labels	0.444187
Paper	0.433918
Envelopes	0.422676
Copiers	0.371956
Fasteners	0.313965

Analysis:

The Sub-Categories are ranked by their Profit Margin as follows:

1. **Labels** — 0.444187
2. **Paper** — 0.433918
3. **Envelopes** — 0.422676
4. **Copiers** — 0.371956
5. **Fasteners** — 0.313965

Chart

Tables

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Sample - Superstore.csv	9,994	21

8. Drawbacks

- **Deterministic vs. Probabilistic Gap:** While the code execution is deterministic, the reasoning model's interpretation of large data aggregates can occasionally require a second 'check' pass.
- **Complexity Overhead:** The multi-stage agentic workflow introduces higher latency compared to a single-model query approach.
- **Dependency on Python Runtime:** The system relies on the underlying Pandas library; complex logical queries that fall outside of pandas-native functions require custom extension.

9. Key technical learnings

- 1. Orchestration over Monoliths:** I learned that complex tasks are best solved by "composition" rather than scaling model size. Smaller, specialized models (Reasoning + Coding) acting in concert significantly outperform a single, larger generalized model on domain-specific constraints.
- 2. The Necessity of Semantic Bridges:** I discovered that LLMs often suffer from "data blindness." Creating human-readable bridges (like Markdown serialization) is a critical engineering step to ensure that the Reasoning model can effectively interpret the output of the Code model.
- 3. Defensive Systems Engineering:** Reliable AI isn't just about prompt engineering; it is about building defensive "sandboxes." By implementing autonomous retry loops and explicit data-type validation, we shift from "hoping the model is right" to "ensuring the system is resilient."

10. Implementation summary

Core Components Deployed:

Dual-Model Orchestration: Phi-4 (Reasoning) + Qwen 2.5 (Coding) via Ollama.

Markdown Bridge: Automated conversion of all data outputs into strict Markdown table formats.

Self-Correction Loop: Autonomous 3-attempt retry logic for failed Python execution.

Defensive Data Pipeline: Automated cleaning, type-checking, and schema-based table identification.

Privacy First: 100% offline, local weights execution with zero external data telemetry.